

Vector Spaces, Linear Transformations, and LU Factorization in Infectious Disease Modeling

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1. What Is Compartmental Disease Modeling and Why Does It Matter?

Infectious disease modeling is one of the most practically significant subfields of applied mathematics. When a new pathogen emerges, public health authorities must answer a deceptively simple question: *will this disease spread through the population, or will it die out?* Epidemiologists could not answer this question without the mathematical foundations that lie in the linear algebra concepts we have studied in Modules 1–5: **vector spaces**, **linear transformations**, **matrix inverses solved via LU factorization**, and **elementary row operations**. With my background in public health modeling, I have decided to dedicate this application paper to “pull back the curtain” and truly understand the origins of one of the most used methods in infectious disease modeling that I personally use in my own work in Tuberculosis modeling.

The dominant modeling framework is the *compartmental model*, which divides a population into mutually exclusive subgroups (“compartments”) and tracks the flow of individuals between them over time. The canonical example is the SIR model, which partitions every person into one of three states:

- S — **Susceptible**: individuals who have not yet been infected and can acquire the disease.
- I — **Infectious**: individuals who are currently infected and can transmit the pathogen.
- R — **Recovered** (or Removed): individuals who have cleared the infection and are immune.

More realistic diseases require more compartments. Tuberculosis, for instance, features a long latency period where a person harbors the bacterium without being infectious; this necessitates an Exposed (E) state, giving the **SEIR** model. A TB cascade-of-care model might further subdivide populations by treatment status, loss to follow-up (LTFU), and drug

resistance, producing state spaces with ten or more compartments.

The central question in each of these models is captured by a single number: the **basic reproduction number** R_0 . Informally, R_0 is the average number of new infections caused by a single infected individual introduced into a fully susceptible population [2]. The threshold criterion is as follows:

if $R_0 < 1$, the disease dies out; if $R_0 > 1$, the disease spreads.

The matrix theory concepts that underly the computing of R_0 in structured populations is the **next-generation matrix** (NGM), introduced formally by Diekmann, Heesterbeek, and Metz in 1990 [1] and systematized for compartmental ODE models by van den Driessche and Watmough in 2002 [3]. Constructing the NGM requires us to view the state space of infected individuals as a **vector space**, decompose the disease dynamics into two **linear transformations** (F and V), and invert V using **LU factorization**. In staged disease models, this inversion reduces to simply forward substitution on a lower-triangular system. The product $K = FV^{-1}$ is the NGM, a linear transformation whose entries encode the expected number of new infections each compartment generates.

New terms introduced in this paper: **compartmental model**, **disease-free equilibrium** (DFE), **next-generation matrix** (NGM), **basic reproduction number** R_0 , **transmission matrix** F , **transition matrix** V , **M-matrix** (a square matrix with non-positive off-diagonal entries whose determinant is strictly positive, guaranteeing invertibility), and **Z sign pattern** (a matrix whose off-diagonal entries are all non-positive).

2. The Mathematical Framework

2.1 The State Space as a Vector Space

Let a disease model have n total compartments. We represent the population's state at time t as a vector $\mathbf{x}(t) = (x_1, x_2, \dots, x_n)^T \in \mathbb{R}_{\geq 0}^n$. The first m components correspond to *infected* compartments (e.g., E and I in an SEIR model); the remaining $n - m$ components are *uninfected* (e.g., S and R). This infected subspace $\mathbb{R}^m \subset \mathbb{R}^n$ is the vector space on which the NGM acts as a linear transformation. The fact that $\mathbf{x}(t)$ lives in $\mathbb{R}_{\geq 0}^n$ reflects the real-world biological constraint (no negative people) that imposes important structural properties on F and V .

The dynamics of the full system are governed by a system of ODEs:

$$\dot{x}_i = \mathcal{F}_i(\mathbf{x}) - \mathcal{V}_i(\mathbf{x}), \quad i = 1, \dots, n,$$

where $\mathcal{F}_i(\mathbf{x})$ is the rate at which *new infections* enter compartment i , and $\mathcal{V}_i(\mathbf{x})$ captures all other flows: progression through disease stages, recovery, death, and movement in and out via treatment [3].

2.2 Linearization at the Disease-Free Equilibrium

The **disease-free equilibrium (DFE)**, denoted $\mathbf{x}^0$, is the point where all infected compartments are empty. To study whether a small introduction of infectious individuals will grow or decay, we linearize the infected subsystem at $\mathbf{x}^0$ by taking partial derivatives. The resulting Jacobian of the infected subsystem decomposes as:

$$\left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\mathbf{x}^0, \text{infected}} = F - V,$$

where $F = \left[\frac{\partial \mathcal{F}_i}{\partial x_j}(\mathbf{x}^0) \right]$ and $V = \left[\frac{\partial \mathcal{V}_i}{\partial x_j}(\mathbf{x}^0) \right]$ are both $m \times m$ matrices [3]. Their key structural properties are:

- F is **non-negative**: every entry is a rate of new infection production, which must be ≥ 0 by biology.
- V is an **M-matrix**: it has non-positive off-diagonal entries (inflows from other infected classes) and positive diagonal entries (total outflows from each compartment). The conditions of the model force $\det(V) > 0$, so V is **invertible**, and moreover $V^{-1} \geq 0$ entry-wise [3].

These two matrices are **linear transformations** on $\mathbb{R}^m$. The transformation F maps a current distribution of infected individuals to the rate of new infections it generates; V maps that same distribution to the rate at which individuals transition out of infected states. Separating these two processes is the conceptual core of the NGM method [2].

2.3 V^{-1} as a Linear Transformation: Time-in-Compartment

Before computing $K = FV^{-1}$, we should understand what V^{-1} means as a linear transformation, because its epidemiological interpretation is what makes the NGM readable.

With reinfection turned off, the infected subsystem evolves as $\dot{\mathbf{w}} = -V\mathbf{w}$, whose solution is $\mathbf{w}(t) = e^{-Vt}\mathbf{w}(0)$. Integrating from $t = 0$ to ∞ gives the total time each compartment is occupied:

$$\int_0^\infty e^{-Vt} dt = V^{-1}.$$

This means the (j, k) entry of V^{-1} is the **expected total time** an individual introduced into compartment k spends in compartment j over their entire infectious career, barring reinfection [3]. Combined with the fact that the (i, j) entry of F is the **rate** at which an individual in compartment j generates new infections in compartment i , the (i, k) entry of

$K = FV^{-1}$ is:

$$K_{ik} = \text{expected number of new infections in compartment } i \\ \text{generated by one individual seeded in compartment } k.$$

In other words, $K = FV^{-1}$ is a linear transformation that maps a vector of initial infection seeds to the vector of infections in the next generation [1].

3. Illustrative Example: The TB Treatment Model

To make the framework concrete, we work through the SEIR-with-treatment model from van den Driessche and Watmough [3], which directly mirrors the TB models I work with. The population is divided into four compartments: S (susceptible), E (exposed/latent), I (infectious), and T (treated), with infected compartments E and I giving $m = 2$. The parameters are: β_1 (transmission rate), ν (progression rate from E to I), r_1 and r_2 (treatment rates for E and I respectively), $p = 1 - q$ (treatment failure probability), and d (natural death rate).

Evaluating the partial derivatives at the DFE (with $S_0 = 1$) gives:

$$F = \begin{pmatrix} 0 & \beta_1 \\ 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} d + \nu + r_1 & -pr_2 \\ -\nu & d + r_2 \end{pmatrix}.$$

The zero in the $(2, 1)$ position of F reflects that newly infected individuals enter E only: no one is placed directly in I upon infection. This means the only **state-at-infection** is E [2], a structural observation that falls directly out of examining which rows of F are nonzero.

Step 1: Compute V^{-1} via row reduction.

We augment V with the identity and apply elementary row operations:

$$\left[\begin{array}{cc|cc} d + \nu + r_1 & -pr_2 & 1 & 0 \\ -\nu & d + r_2 & 0 & 1 \end{array} \right].$$

Let $\delta = \det(V) = (d + \nu + r_1)(d + r_2) - \nu pr_2$. Since V is an M-matrix, $\delta > 0$ [3]. Row-reducing to the identity on the left yields:

$$V^{-1} = \frac{1}{\delta} \begin{pmatrix} d + r_2 & pr_2 \\ \nu & d + \nu + r_1 \end{pmatrix}.$$

Every entry of V^{-1} is non-negative (since all parameters are positive and $\delta > 0$), confirming the time-in-compartment interpretation.

Step 2: Form the NGM $K = FV^{-1}$.

$$K = \begin{pmatrix} 0 & \beta_1 \\ 0 & 0 \end{pmatrix} \cdot \frac{1}{\delta} \begin{pmatrix} d + r_2 & pr_2 \\ \nu & d + \nu + r_1 \end{pmatrix} = \frac{1}{\delta} \begin{pmatrix} \beta_1 \nu & \beta_1(d + \nu + r_1) \\ 0 & 0 \end{pmatrix}.$$

Step 3: Read R_0 from the structure of K .

The second row of K is zero, reflecting the fact that I is not a state-at-infection: the I compartment produces infections, but no new infections are seeded directly into I . Row-reducing K (pivoting on the $(1, 1)$ entry, which is the only nonzero pivot) shows that K has rank 1. When K has rank 1, the NGM generates exactly one independent infection pathway, and R_0 is the unique nonzero entry in the row-echelon form of K :

$$R_0 = \frac{\beta_1 \nu}{\delta} = \frac{\beta_1 \nu}{(d + \nu + r_1)(d + r_2) - \nu pr_2}.$$

Interpretation. The numerator $\beta_1 \nu$ is the product of the transmission rate and the rate

at which latent individuals progress to infectiousness. The denominator δ encodes the total competition between disease progression and all forms of outflow (death, treatment, treatment failure cycling). As treatment improves through larger r_1 , r_2 , or q , δ increases, R_0 decreases, and the linear transformation K shrinks the next generation. In other words, better treatment reduces epidemic potential.

4. Advanced Example: LU Factorization and the Staged Progression Model

4.1 Structure of the Staged Progression Model

The TB treatment example above used a 2×2 matrix V that we inverted by row reduction. In a **staged progression model**, infected individuals move sequentially through m disease stages with changing infectivity. This framework applies directly to HIV, TB drug-resistance progression, and multi-stage cascade-of-care models. In this setting, the matrix V takes on a lower-bidiagonal structure that is solved by forward substitution, which is the solve step of **LU factorization** with $L = V$ and $U = I$.

The model has a single susceptible compartment S and m infectious stages $I_1, I_2, \dots, I_m$, with progression rate ν_i out of stage i and disease-induced death rate d_i at stage i [3]. The infected subsystem linearized at the DFE gives:

$$F = \begin{pmatrix} \beta_1 & \beta_2 & \cdots & \beta_{m-1} & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} \nu_1 + d_1 & 0 & 0 & \cdots & 0 \\ -\nu_1 & \nu_2 + d_2 & 0 & \cdots & 0 \\ 0 & -\nu_2 & \nu_3 + d_3 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & -\nu_{m-1} & d_m \end{pmatrix},$$

where the final stage I_m has zero infectivity ($\beta_m = 0$) due to morbidity, and $\nu_m := 0$ is set

for notational convenience [3]. Because V is already lower triangular, it is its own L factor in an LU decomposition with $U = I$.

4.2 Lemma: F is Non-Negative and V is an Invertible M-Matrix

The following result from van den Driessche and Watmough [3] (Lemma 1) establishes the two structural properties that make the NGM well-defined. All of the epidemiological assumptions cited below are stated formally in that paper; here we describe them in plain terms.

Theorem 1 (Lemma 1, van den Driessche & Watmough [3]). *Let $\mathbf{x}^0$ be a DFE of the compartmental model. Under the biological assumptions of the model (non-negativity of population counts, no immigration of infectives, and stability of the DFE in the absence of infection), the matrices F and V defined by the linearization at $\mathbf{x}^0$ satisfy: (i) $F \geq 0$ entry-wise, (ii) V is a nonsingular M-matrix, and (iii) $V^{-1} \geq 0$ entry-wise.*

Proof summary, following van den Driessche & Watmough [3]. We summarize the three conclusions in turn.

(i) $F \geq 0$. The function $\mathcal{F}_i(\mathbf{x})$ counts new infections entering compartment i , so it is non-negative for all $\mathbf{x} \geq 0$ by biology. At the DFE, the biological assumptions also require $\mathcal{F}_i(\mathbf{x}^0) = 0$ for infected compartments (since there is no disease present to generate new infections). The entry $F_{ij} = \frac{\partial \mathcal{F}_i}{\partial x_j}(\mathbf{x}^0)$ is a one-sided limit of the form $\lim_{h \rightarrow 0^+} [\mathcal{F}_i(\mathbf{x}^0 + h\mathbf{e}_j) - \mathcal{F}_i(\mathbf{x}^0)]/h$. Because $\mathcal{F}_i \geq 0$ and $\mathcal{F}_i(\mathbf{x}^0) = 0$, this limit is ≥ 0 . Thus $F \geq 0$.

(ii) V is a nonsingular M-matrix. We show V has the Z sign pattern and positive diagonal, which together with the stability assumption imply V is a nonsingular M-matrix. For the off-diagonal entries: if compartment i is empty ($x_i = 0$), no individuals can leave it, so $\mathcal{V}_i^-(\mathbf{x}) = 0$. The entry V_{ij} for $i \neq j$ is the partial derivative of net outflow from compartment i with respect to x_j . Because $x_i = 0$ forces $\mathcal{V}_i^-(\mathbf{x}^0 + h\mathbf{e}_j) \leq 0$, we get $V_{ij} \leq 0$

for $i \neq j$, giving the Z sign pattern. The biological assumption that the DFE is stable when $F = 0$ forces all eigenvalues of $-V$ to have negative real parts, which is equivalent to V having all eigenvalues with positive real parts. A matrix with the Z sign pattern and positive real-part eigenvalues is by definition a nonsingular M-matrix, so $\det(V) > 0$ and V is invertible.

(iii) $V^{-1} \geq 0$. A standard result in the theory of M-matrices states that if V is a nonsingular M-matrix, then $V^{-1} \geq 0$ entry-wise [3]. Intuitively, this is consistent with the time-in-compartment interpretation: expected times are non-negative.

This lemma is the structural foundation of the entire NGM framework. Once we know $F \geq 0$, $V^{-1} \geq 0$, and $\det(V) > 0$, the product $K = FV^{-1}$ is well-defined and non-negative, and its entries carry the infection-generation interpretation described in Section 2.3. $\square$

4.3 Reading R_0 from $K = FV^{-1}$

Since only the first row of F is nonzero, $K = FV^{-1}$ also has only a nonzero first row, and R_0 is the sum of the first-row entries of K . Using the closed-form entries of V^{-1} that follow from forward substitution on the lower-triangular V [3] (equation 12 therein):

$$R_0 = \sum_{j=1}^{m-1} \beta_j \cdot (V^{-1})_{1j} = \frac{\beta_1}{\nu_1 + d_1} + \frac{\beta_2 \nu_1}{(\nu_1 + d_1)(\nu_2 + d_2)} + \cdots + \frac{\beta_{m-1} \prod_{k=1}^{m-2} \nu_k}{\prod_{k=1}^{m-1} (\nu_k + d_k)}.$$

Each term is transparent: the j -th term is the product of β_j (infectivity in stage j) and the fraction of individuals originally seeded in stage 1 who survive to reach stage j , multiplied by the time they spend there. LU factorization via forward substitution on a lower-triangular V delivers this interpretation directly.

In a TB cascade-of-care model, adding more stages (diagnosed, on-treatment, LTFU, re-treated) extends the size of V but preserves its lower-bidiagonal structure. Each new stage appends one row and one column, and forward substitution remains the efficient path to V^{-1} .

The NGM makes each intervention lever visible as a distinct entry in a linear transformation, giving both modelers and policymakers a precise handle on which parameters most influence R_0 . After learning about and using compartmental models in public health research for the past 3 years, it is incredible to finally understand how the methods were developed and the matrix theory concepts that made it possible.

References

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